

COMPX221 Practical Test 1

2021

First Name/Given Name	Username
Surname/Family Name	ID Number

Assessment

10	AbstractObject class implemented correctly 1 mark for abstract class and 2 marks for fields 3 marks for constructor 3 marks for abstract methods 1 mark for toString() method
10	WaterCatcher class implemented correctly 1 mark for inheritance and 1 mark for fields 3 marks for constructor 2 marks for toString() method 2 marks for drawObject() method 1 mark for growObject() method
10	RainDrop class implemented correctly 1 mark for inheritance and 1 mark for fields 3 marks for constructor 2 marks for toString() method 1 marks for drawObject() method 1 mark for growObject() method 1 mark for moveObject method
10	Loading the csv file of data correctly and adding objects to the list
2	Info about all objects displayed in console window
2	Drawing all objects to the sketch window
2	Rain drops will move down the sketch window
2	New objects can be added using the mouse buttons
2	Random objects will grow in size
	Total

Getting Started

To begin, download the zip file of the template folder from Moodle and unzip the file. Then add the word Answer to the end of the folder name. Go into the folder and into the Prac Test folder and then open the sketch in processing. At the end of the test you need to zip up the folder that has the word answer in the name and submit that to the Prac Test 1 Submission activity and make sure you click the submit button.

Test Conditions

Practical Test 1 will be an open book test. You can refer to any printed notes, look at information on Moodle as well as use the internet. You have 105 minutes to complete as much as you can.

You are not allowed to use any of the following during the test:

- email applications or file sharing applications
- facebook or other social media websites, online chat applications
- Any other applications which allows communication with other people.
- tablets, phones, music players
- USB drives

If you have managed to get the answers for this test, then you are not allowed to refer to them during the test.

Anyone found guilty of breaching these conditions will get sent to the University Discipline Committee.

Background

You have been asked to help work on a game for small children to help them learn about the importance of global warming. The earth has become desolate and rains very infrequently so when rain is falling the remaining population must collect as much water as they can. Nileshe has asked you to model the rain drops and water catchers as he thinks about the mechanics of the game. Take a look at the class diagram provided inside the template sketch folder.

Some code in the sketch tab is provided for you test your classes before processing the CSV file of data. You should just comment and uncomment the code as required when testing.

A csv file has been provided for you called objects.csv. It has 2 different types of lines in the file.

A water catcher line has the following format (all values are integers):

`x,y,width,height`

A rain drop line has the following format (all values are integers):

`x,y,size`

Tasks

1. Implement the `AbstractObject` class (which is an abstract class) as specified in the class diagram using the notes below:

The constructor will create a new `PVector` object for the position using the `x` and `y` values passed in and set the `objectColor` field to the value passed in.

The `drawObject()`, `moveObject()` and `growObject()` methods are all abstract methods.

The `toString()` method returns back the `x` and `y` position as a string with a comma between both values (i.e. `30,50`).

2. Implement the `WaterCatcher` class as specified in the class diagram using the notes below:

The constructor will set the position to the `x` and `y` values passed in and also set the colour to `#B0BF13`. Think carefully how to do this so you don't repeat any code. Then also set the width and height of the water catcher to the values passed in.

The `toString()` method will return the word "Catcher:" as well as the `x` and `y` position and the width and height of the water catcher to the console window in the following format:

`Catcher: x,y width height`

Note: Think carefully how to write the code so that there is no repeated code.

The `drawObject()` method should draw a rectangle for the water catcher centred around its `x` and `y` position, in the correct width and height and colour.

The `moveObject()` method should be created but leave it empty as Nilesh hasn't figured out what type of movement the water catcher will have.

The `growObject()` method should make the width of the water catcher increase by the amount passed to the method.

Test your class by creating and displaying some water catchers by uncommenting the provided code in the sketch tab. Once they are displaying correctly comment out the code again.

3. Implement the RainDrop class as specified in the class diagram using the notes below:

The constructor will set the position to the x and y values passed in and also set the colour to #1370BF. Think carefully how to do this so you don't repeat any code. Then also set the size of the rain drop to the value passed in.

The toString() method will return the word "RainDrop:" as well as the x and y position and the size of the rain drop to the console window in the following format:

```
RainDrop: x,y size
```

Note: Think carefully how to write the code so that there is no repeated code.

The drawObject() method should draw a circle for the rain drop centred around it's x and y position, in the correct size and colour.

The moveObject() method should move the rain drop down the sketch window by the amount passed in.

The growObject() method should make the size of the rain drop increase by the amount passed to the method.

Test your class by creating and displaying some Rain Drops by uncommenting the provided code in the sketch tab. Once they are displaying correctly comment out the code again.

4. Create an array list at class scope level named objectList in the sketch tab. Think carefully what the datatype of the array list should be so that you can store objects from both subclasses in the list.
5. In the setup method write the code to load all the data from the csv file and create the appropriate objects from the sub classes and store all of the objects in the objectList.
6. After loading the data from the csv file display information from each object to the console window.

Note: Don't worry if the x and y values are displayed float values.

7. In the draw method in the sketch tab, write the code that will draw all of the objects to the sketch window. Test that all of the objects are displayed. Then modify the code so that after drawing the object it is moved by 1 pixel. Test that your objects now move (the water catchers won't move yet).
8. In the sketch tab, create the mousePressed() method and write the code so that if the left button is pressed a new water catcher is created at the mouse position with a width of 50 and a height of 10 and is added to the object list.

If the right mouse button is pressed then write the code to create a new rain drop at the mouse position with a size of 20 and is added to the object list. Test that you can now create new objects when pressing the mouse buttons.
9. At the bottom of the draw() method in the sketch tab, write the code to randomly make an object from the object list grow by 1 pixel.